

BERKE TAYFUN AKSEKI

AI / ML Engineering | Data Analytics | Backend Development

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PROFILE

Computer Science and Management graduate from LUISS Guido Carli, Rome (Jul 2026). Shipped 7 end-to-end projects, all public on GitHub: a production RAG pipeline (sentence-transformer embeddings + FAISS vector search), ML APIs, an A/B-testing & causal-inference toolkit, analytical SQL, a JWT-authenticated job-market tracker, a multiplayer game, and a live portfolio site — most containerised with Docker and tested in CI. Combines Python / SQL engineering depth with business strategy from a dual-track degree. Targeting junior roles in AI / ML Engineering, Data Analytics, or Backend Development; open to relocation across Europe.

EDUCATION

LUISS Guido Carli University — Rome, Italy

Sep 2023 – Jul 2026

Bachelor's Degree in Management and Computer Science (L-18)

- Thesis: "Tourism Dependence and Regional Macroeconomic Resilience in Italy" — continuous-treatment difference-in-differences with two-way fixed effects and event-study analysis across 21 NUTS 2 regions over a 10-year panel (2015–2024), Eurostat / ISTAT data in R; identified structural post-pandemic labour-market hysteresis in high-tourism regions. Supervisor: Prof. Diletta Topazio.
- Top coursework (out of 30): Statistics 30, Finance & Financial Technologies 29, Microeconomics 27, Management of Technology 27, Algorithms 25.

TECHNICAL SKILLS

Languages & Frameworks: Python (Pandas, NumPy, scikit-learn, SciPy, statsmodels), SQL, R, FastAPI, Pydantic, SQLAlchemy, Django, C#, JavaScript, HTML / CSS

AI & ML Engineering: RAG (Retrieval-Augmented Generation), embeddings, semantic search, vector search (FAISS), sentence-transformers / Hugging Face, PyTorch, TensorFlow / Keras, NLP, TF-IDF, cosine similarity, document chunking, ML pipelines, model serving (REST)

Computer Vision & Multimodal: OpenCV, computer vision, image classification, object detection, CNNs, video processing & frame extraction (FFmpeg), audio / speech processing, Whisper (speech-to-text), automated content tagging, feature extraction

Data & Analytics: Power BI, KNIME, Streamlit, Matplotlib, Excel, ETL, feature engineering, model evaluation, KPI dashboards, A/B testing, causal inference (DiD), hypothesis testing, econometrics

Tools & DevOps: Git / GitHub, Docker, GitHub Actions (CI/CD), pytest, PostgreSQL, SQLite, REST APIs, JWT / OAuth2 auth, BeautifulSoup, Unity 6, Blender, Jupyter, VS Code, RStudio

SELECTED PROJECTS

Portfolio Website — berketayfunakseki.com (Live) — HTML · CSS · JavaScript · GitHub Pages — berketayfunakseki.com

- Zero-dependency portfolio with an interactive JD-Match tool (recruiters paste a job description; client-side TF-IDF scores my profile live) and a Recruiter / Engineering toggle exposing per-project architecture case studies.

Causal Lab — A/B Testing & Difference-in-Differences Toolkit — Python · SciPy · statsmodels · FastAPI · Streamlit — github.com/berketayfunakseki/causal-lab

- Statistics-first experimentation toolkit: two-proportion z-tests with confidence intervals, sample-size / power analysis, minimum-detectable-effect and test-duration estimators, automated sample-ratio-mismatch detection, and a two-way fixed-effects difference-in-differences estimator with clustered standard errors — 7 passing pytest cases each verify a method recovers a known injected effect from simulated data.

SQL Analytics Lab — Cohort Retention & Funnel Analysis — SQL (window functions, CTEs) · Python · Streamlit — github.com/berketayfunakseki/sql-analytics-lab

- Production-style analytical SQL over a synthetic app-usage dataset: weekly cohort retention, step-over-step funnel conversion (LAG, FIRST_VALUE), growth metrics (running totals, RANK vs DENSE_RANK) — surfaced the steepest funnel drop-off directly from the data.

Labyscape — Multiplayer Co-op Horror Game (In Development, repo private) — Unity 6 · C# · Netcode for GameObjects · Blender

- Architected a 2–8 player co-op horror game: procedural 18×18 DFS maze, host-authoritative Netcode multiplayer, NavMesh enemy AI, health / respawn systems in Unity 6 URP — sole developer, targeting Steam.

Jobscope — Swiss / EU Job Market Tracker — FastAPI · SQLite · scikit-learn · Docker — github.com/berketayfunakseki/jobscope

- Automated ETL pipeline aggregating 200+ live listings from RemoteOK and Adzuna APIs; deduplicates by normalised URL and scores each role 0–100 against a candidate profile via TF-IDF cosine similarity; FastAPI backend with JWT auth (python-jose, bcrypt), SQLAlchemy ORM (SQLite, Postgres-swappable), containerised with Docker.

Regulatory RAG Assistant — sentence-transformers · FAISS · FastAPI · Docker · CI — github.com/berketayfunakseki/regulatory-rag-assistant

- End-to-end Retrieval-Augmented Generation pipeline over regulatory PDFs: pdfplumber ingestion → overlapping chunking → sentence-transformer embeddings (all-MiniLM-L6-v2) → normalised FAISS cosine-similarity index → top-k semantic retrieval, served via FastAPI REST + Streamlit; Dockerised with pytest running in GitHub Actions CI.

Churn Prediction — ML API + Dashboard — scikit-learn · FastAPI · Streamlit

- Trained a customer-churn classifier with reproducible scikit-learn pipelines; served predictions via FastAPI endpoint; visualised model outputs in a Streamlit dashboard.

EXPERIENCE

LUISS Language Cafe — LUISS Guido Carli University

Feb 2026 – May 2026

Creative Manager Intern • Rome, Italy

- Owned creative direction for the university language-exchange programme: social graphics, newsletters, and event copy across LUISS channels.
- Launched multi-format campaigns aligned with brand identity, increasing programme visibility and event attendance.

LANGUAGES

Turkish (native) • English (C1 — full professional) • French (B2) • Italian (B1)